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New strategies for the synthesis of ZnO and Al-doped ZnO films by reactive magnetron sputtering at room temperature

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We review our recent findings that reactive magnetron sputtering can be operated without thermal assistance to produce epitaxial ZnO films and highly conductive and transparent aluminum-doped ZnO (AZO) films on large surface area. The growth of epitaxial films requires working in direct current (DC) mode at large oxygen partial pressure but high power impulse magnetron sputtering (HiPIMS) is preferred to optimize the electrical and optical properties of AZO films on large surface areas. The results are interpreted by the ability of DC sputtering to easily oxidize the target surface and the HiPIMS mode cleans and returns it to metallic during pulses. Fast oxygen atoms (ions) may be emitted in large amount using DC and bring fuel to the growing ZnO film to induce epitaxy onto sapphire. The signature of oxygen interstitials is found in DC with increasing magnitude as the oxygen content is increased in the gas phase. In contrast, the discharge voltage plays a significant role in HiPIMS to adjust the sputtering rate and clean the target surface for deposition of slightly sub-stoichiometric AZO films over the entire range of lateral positions. Thereby, transparent yet highly conductive films with resistivity in the range 4-15 10^{-4} Ωcm can be produced.

1 Introduction Zinc oxide (ZnO) is one of the most studied materials due to a unique set of functional properties. It is a wide bandgap semiconductor with dopability-related simultaneous large electronic conductivity, optical transparency in the visible in infrared regions. Its ability to emit light in the UV and visible at room temperature following optical or electrical stimulations, due to its large exciton binding energy (60 meV), is also highly considered. The highest potential of applications of ZnO is probably carried by optical and electrical properties for use as luminescent layer in UV and visible LEDs and lasers, as transparent electrodes for electrochromic devices, photovoltaic solar cells or flat panel displays. Moreover, ZnO is earth-abundant, non-toxic and a low cost material. It is therefore considered as one of the most serious alternatives to indium tin oxide (ITO) transparent electrodes. Many strategies are applied to synthesize ZnO in different sizes and shapes and for doping it in view of specific applications. Synthesis methods involving wet chemistry are increasingly considered due to low associated investment costs and possibility to easily produce nanostructures. Nevertheless, it is sometimes difficult to avoid the presence of contaminants, limiting the possibilities to optimize the properties. Physical or chemical vapor deposition methods, though requiring higher initial financial investments, can produce films of high purity and present the advantage to be compatible with the standard methods of the microelectronic industry, a strong advantage if device fabrication is targeted. Here we focus on reactive magnetron sputtering, a physical vapour deposition method well-established both in laboratories and industry for the synthesis of compound thin films.

It is generally believed, based on experimental observations, that optimization of the optical and electrical properties of ZnO films synthesized by magnetron sputtering requires substantial thermal assistance. For instance, large photoluminescence (PL) in the UV, due to recombination of excitons, and low intensity in the visible range are generally interpreted as a consequence of good crystallinity and limited density of point defects in ZnO. Vapor condensation may be associated to cooling rates as high as 10^{15} Ks^{-1}, keen to produce far out of stable equilibrium structural and miscrostructural states and large density of point defects, supposedly detrimental to UV PL. Nevertheless, this might be a limited view as there is no comprehensive understanding on the influence of point defects on PL properties of ZnO films and the interpretation of luminescence spectra is still highly debated. Similarly, high crystallinity in

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polycrystalline or monocrystalline epitaxial films is perceived as the ideal structural state to set large conductivity in transparent ZnO films and as only achievable by substantial thermal assistance during growth. Nevertheless, many factors affect the conductivity of wide bandgap semiconductors, such as the amount, activation and compensation of dopants as well as electron scattering mechanisms [1]. There are several motivations to avoid or limit thermal assistance during sputter-deposition of ZnO. There is strong interest to ease the synthesis process and lower the power consumption. Moreover, some materials to be interfaced with ZnO in devices cannot withstand high temperatures. This is for instance the case of copper indium gallium selenide (CIGS) absorbers in CIGS solar cells that tend to degrade above 200°C [2]. Another strong motivation is the rapid development of electronics, whether or not transparent, on flexible (often organic) supports for which ZnO is a serious candidate to replace indium tin oxide (ITO) transparent electrodes.

In section 2, we review on two of our recent findings related to the control of the microstructure and properties of zinc oxide films. The first part is dedicated to the description of the characteristics and possible origin of near-room temperature single domain epitaxy in such films deposited by reactive magnetron sputtering. The second part discusses the limitations of conventional "room temperature" reactive magnetron sputtering of ZnO transparent electrodes and highlights the possibility to produce highly conductive and transparent Al-doped ZnO (AZO) on large surface areas using high power impulse magnetron sputtering (HiPIMS) without thermal assistance.

All results were obtained using laboratory scale deposition chambers with the target eccentric to a rotating substrate holder. This enables to deposit coatings with little variations in the compositions of sputtered atoms [3] and tends to amplify the inhomogeneity of functional properties compared to in-line industrial chambers in which they are averaged.

2 "Room temperature" synthesis of ZnO films.

2.1 Epitaxial growth Because it prevents dangling bonds, charge carriers trapping and scattering at the grain boundaries that are common in polycrystalline materials [4, 5], epitaxial ZnO films are of strong interest for optimized electronic and optical properties. Various methods have been reported to lead to epitaxial ZnO thin films such as metalorganic vapour phase epitaxy (MOVPE) [6], molecular beam epitaxy (MBE) [7] or pulsed laser deposition (PLD) [8, 9]. A common feature of such methods is the use of substantial thermal assistance above 300 ◦C, in most cases above 500°C, during growth. Nevertheless, virtually, epitaxy may be possible at any temperature, tough the thickness of epitaxial breakdown is decreased exponentially with the growth temperature during molecular beam homoepitaxy of Ge(001) [10]. Interestingly, Shin and Aziz showed that higher kinetic energy of particles strongly delays the epitaxial breakdown [11]. Low temperature heteroepitaxy of reactively sputtered ZnO on sapphire was reported in 2004 by Ruthe et al. [12] with a measured self-established temperature of ~80°C by contacting a thermocouple with the substrate after deposition, indicating probably higher temperature at the film during deposition. Similar results were later reported by Vinnichenko et al. [13]. Kinetic energy of sputtered species and oxygen partial pressure were suspected by the authors to play a significant role in the possibility to establish such growth mode.

We studied the possibility to establish epitaxial growth, the film optical and structural properties while varying the oxygen content in the gas in the range 4.5 – 11.5 10^{-2} Pa of the oxygen partial pressure [14]. All films were oriented with the c-axis of the wurtzite cell of perpendicular to the film surface. Figure 1 shows X-ray diffraction φ scans of 150 nm-thick films.

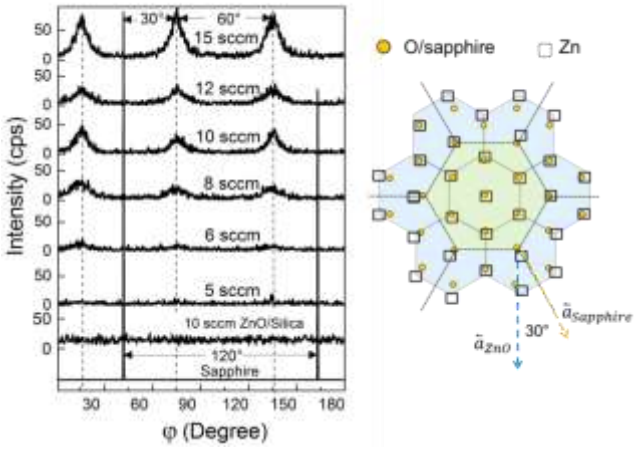


Figure 1 Left, XRD φ-scans for $\{11\bar{2}2\}$ planes in ZnO films deposited on (0 0 0 1)-sapphire for different oxygen flow rates. The signals from the $\{10\bar{1}4\}$ planes of sapphire and from a ZnO film deposited on silica are used as references (reprint from [14], copyright IOP). Right, schematic description for the epitaxial orientation between ZnO and sapphire in accordance to the φ-scans.

It reveals the the 6-fold symmetry of the ZnO wurtzite structure develops while rising the oxygen flow rate (OFR) above a critical value of 8 standard cubic centimeters per minute (sccm). The temperature measured in situ was lower than 40°C in all cases. The threshold of 8 sccm corresponds to minimization of in-plane internal mechanical stresses, which can be interpreted as the condition to produce the lowest density of interstitial point defects and composition closest to oxygen stoichiometry [14]. Analysis of the PL signal tends to confirm this hypothesis (Figure 2). The signal was deconvoluted in 3 contributions P1-3.

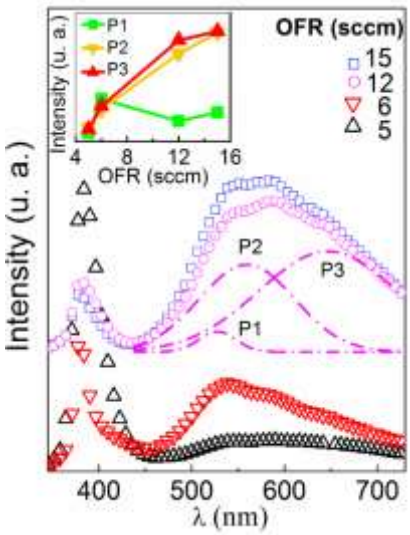


Figure 2 Photoluminescence spectra of ZnO films deposited under different oxygen flow rates (OFR), including an exemple for the deconvolution of the visible signal with 3 contributions. The inset shows the evolution of the maximal intensity of the contributions with the OFR (reprint from [14], copyright IOP).

As mentioned in the introduction, the interpretation of the PL signal of ZnO is highly debated but the correlation of P1 and P3 to the oxygen content [14] led us to rather obvious interpretations. Contribution P1 (maximum at 527 nm), often referred as green contribution, is ascribed to the transition between a shallow level (Zn_i) and an oxygen vacancy (V_O) level. Contribution P2 (560 nm, yellow) was ascribed to a transition between the conduction band and the deep level of oxygen interstitial (O_i). P3 (648 nm orange–red) was not clearly ascribed in the literature but correlated to excess of oxygen [15]. Consistently, it evolves similarly to P1 in our samples. We can interpret the evolution as a progressive change in the dominant point defects from donors at low oxygen partial pressure to acceptors at large oxygen partial pressure. Welzel and Ellmer have clearly evidenced that fast oxygen ions (O^-) are produced during reactive sputtering of ZnO and bombard the film during growth [16]. Figure 3 shows bright and dark field transmission electron microscopy (TEM) images and selected area diffraction (SAED) patterns of a polycrystalline films grown at low oxygen flow rate and a typical epitaxial ZnO film grown above 8 sccm. The bright field image of the polycrystalline film shows a rough surface with maximum amplitude close to 15 nm. In contrast, the maximum amplitude for the epitaxial film is close to 3 nm. This difference is easily explained by considering that species of high kinetic energy can break small clusters of atoms at the film surface [17] and give mobility to surface adatoms [11] to prevent kinetic roughening. The SAED pattern of the epitaxial film is clearly a signature of single crystal ZnO and allows determining the orientation relationships as $ZnO\ [10\bar{1}0]/Al_2O_3\ [11\bar{2}0]$ in plane, with the c-axis of both crystal lattices out of plane.

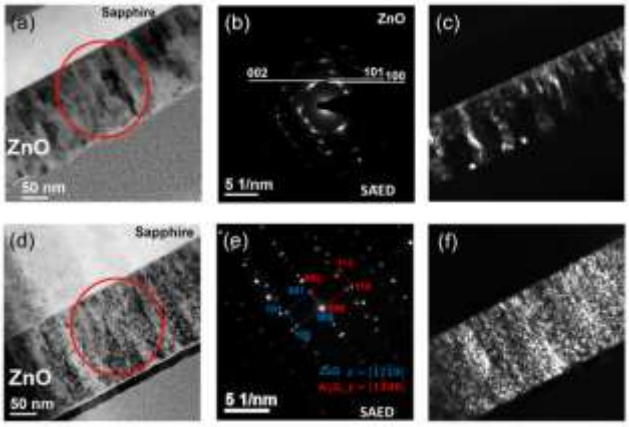


Figure 3 Bright field, dark field images and selected area electron diffraction pattern of ZnO films deposited on (0 0 0 1)-oriented sapphire at OFRs of 5 sccm *(a)–(c)* and 15 sccm *(d)–(f)* (reprint from [14], copyright IOP).

This corresponds to 30◦ of rotation of the a-axis of ZnO with respect to the a-axis of Sapphire as illustrated in figure 1 and already deduced from XRD φ-scans. These orientation relationships lead to the lowest lattice mismatch, around 1.4%.

The possibility to achieve epitaxy at, or close to, room temperature is very likely a consequence of the bombardment by fast O^- particles, consistently with the fact that breakdown of epitaxial growth delayed by the kinetic energy of particles [11]. Epitaxial films grown at high temperature usually must contain few point defects, which relative densities should be closer and closer to the stable equilibrium state, of the growth temperature and oxygen partial pressure, as the temperature is increased. In contrast, room temperature epitaxy offers the possibility to set far out stable equilibrium densities of point defects and vary their nature. This can serve as a platform to study new optical and electrical effects in epitaxial films as well as offers the possibility to separate growth and annealing steps.

2.2 Synthesis of highly conductive and transparent electrodes Zinc oxide, though apparently a material of very simple chemistry, exhibits complicated defect chemistry at the origin of its interesting optical and electrical properties. Even in intrinsic ZnO, several defects occur to give charge carriers that can compensate and establish electronic conductivity. The type of conductivity (n or p) results from the dominant carrier after compensation of electrons and holes. Except in very few cases, the reported intrinsic conductivity of ZnO is generally n-type due to the compensation of holes by electrons [18]. Nevertheless, the conductivity of intrinsic ZnO is generally not sufficient for application as a transparent electrode. Hence, extrinsic doping is often necessary. Many types of heterovalent ions have been considered for doping of ZnO ; Ga^{3+} and Al^{3+} are the most employed. These elements substitute zinc atoms and free one electron upon ionization. In a very simplified picture we can summarize for the case of Al-doped ZnO (AZO), the

effect of intrinsic and extrinsic defects, provided we forget the influence of defect complexes, using the following equation based on neutrality condition:

$$[V_O^{\circ}] + 2[V_O^{\circ\circ}] + [Zn_i^{\circ}] + [Al_{Zn}^{\circ}] + [h^{\circ}] = [V'_{Zn}] + 2[V''_{Zn}] + [O'_i] + [e'] \quad (1)$$

With $[V_O^{\circ}], [V_O^{\circ\circ}], [V'_{Zn}]$ and $[V''_{Zn}]$ corresponding to concentrations of singly and doubly ionized oxygen and zinc vacancies, $[Zn_i^{\circ}]$ and $[O'_i]$ to concentrations of singly ionized zinc and oxygen interstitials, $[Al_{Zn}^{\circ}]$ corresponding to the concentration of aluminum at the zinc (single net charge) and, finally, $[h^{\circ}]$ and $[e']$ to concentrations of free holes and electrons. One sees that ionized oxygen vacancies, zinc interstitials and active aluminum should be present in large concentrations while the presence of oxygen interstitials and zinc vacancies should be limited if large electron conductivity is targeted. In n-type ZnO, the conductivity can be expressed as the product of electron density n_e and mobility μ_e:

$$\sigma = n_e \mu_e \quad (2)$$

Hence, it is also necessary to optimize μ_e. using reactive magnetron sputtering, when thermal assistance is avoided, the condensation occurs in far out of stable equilibrium and the density of point defects is not simply set by the local temperature and oxygen partial pressure during synthesis. Large densities of point defects are expected. Moreover, we have seen in the previous section that fast oxygen particles impinge the growing film. Although this was helpful for the formation of epitaxial ZnO without thermal assistance, it can generate oxygen interstitials and zinc vacancies that compensate the donors [19, 20]. Ellmer and Weltzel have recently reviewed this issue that seems to be generic in transparent conductive oxides [21] and was already reported in 1982 by Minami et al. [22]. It causes local degradations of the electrical conductivity. Moreover, it could also cause damages to other layers interfaced with AZO and may explain the need to use a CdSe buffer layer between the absorber layer and the AZO/i-ZnO bilayer of CIGS solar cells [2]. While it is rather simple to obtain simultaneously high transparency and conductivity for coatings deposited off the magnetron axis, coatings receiving large fluxes from the target erosion track exhibit low conductivity or are insulating depending on the composition of the gas phase [23-25]. Figure 4 illustrates this effect by showing the variation of the electrical resistivity as a function of the sample position for transparent films deposited at two different oxygen flow rates of 5 and 5.5 sccm corresponding to values of the oxygen partial pressure close to 7.5 10^{-2} and 8.25 10^{-2} Pa, respectively.

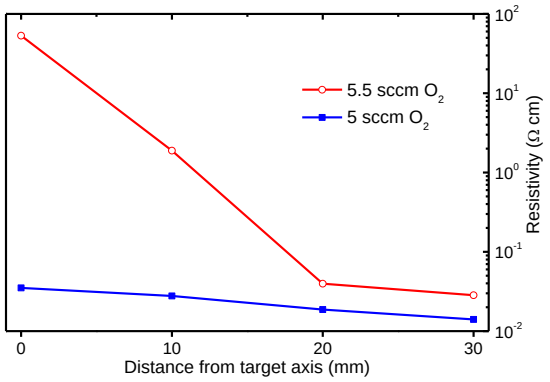


Figure 4 Evolution of the electrical resistivity of AZO films deposited by reactive magnetron sputtering with two OFR (reprint from [25], copyright Elsevier).

It is observed that the resistivity is higher for coatings facing the target axis and this effect amplifies rapidly with the increase in the oxygen content in the gas phase. It can be linked to the non-linear oxidation of the metallic target and is consistent with effects related to the bombardment of the growing film by fast oxygen particles. As commented above, the conventional interpretation of such behaviour is the introduction of acceptor defects compensating the donors. Nevertheless, we have performed electronic structure measurements using X-ray absorption near edge structure at the Al K edge indicating another possible mechanism in action. Figure 5 shows the spectra corresponding to positions 0 and 30 mm in Figure 4 for the OFR of 5 sccm.

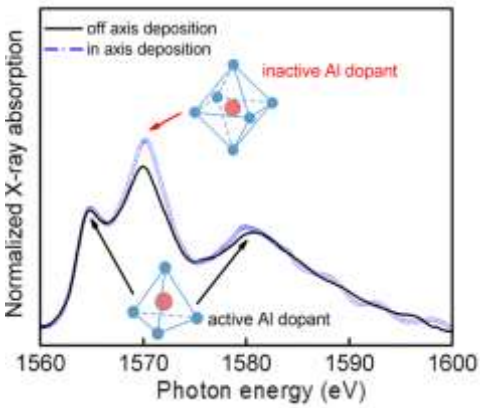


Figure 5 Total fluorescence yield Al K edge XANES spectra of AZO thin films deposited at circumferences of 0 and 30mm from the target axis.

It reveals 3 main contributions. The lowest and highest in energy are attributable to aluminum atoms in tetrahedral conformation with oxygen atoms (coordination number 4 : CN4). It is electrically active. In this conformation, aluminum atoms substitute zinc ones. The intermediate contribution mostly corresponds to an octahedral conformation of aluminum with oxygen atoms (coordination number 6 : CN6). It is associated to excess of oxygen that hinders the possibility to free electrons from aluminum. Both films have similar aluminum contents close to 1.5 at %. Therefore, the

variation in electrical conductivity/resistivity cannot be explained on the basis fluctuations in the aluminum content but is related to differences in dopant activation (see refs 23 and 24 for a more complete discussion).

High Power Impulse Magnetron Sputtering (HiPIMS) has emerged as a new development of magnetron sputtering. HiPIMS is operated at similar average power as conventional sputtering but with concentration of the power in short pulses separated by a long pause time. Several new effects have been observed with HiPIMS but are beyond the scope of the present paper. The relevant effect to this report is instantaneous electrical power density at the target that is larger in HiPIMS than in conventional sputtering. This leads to large instantaneous flux of sputtered atoms that may preserve at least locally the metallic state of the target surface. Reactive HiPIMS has been used by Tiron et al. [26] and Ruske et al. [27] to synthesize transparent AZO films of low electrical resistivity, in the order of few 10^{-3} Ωcm, without thermal assistance. We very recently conducted similar investigation with lateral resolution of the optical and electrical properties in order to evaluate the interest of reactive HiPIMS for the deposition of transparent AZO films highly conductive on large surfaces [28]. Figure 6 shows the evolution of the electrical resistivity with position for various values of the discharge voltage using HiPIMS and a direct current (DC) reactive magnetron sputtering discharge operated at similar averaged power.

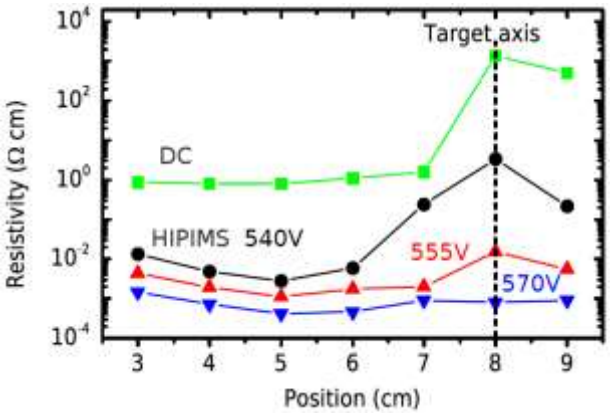


Figure 6 Evolution of the resistivity with the position of samples for different synthesis conditions of AZO films. (reprint from [28], copyright Elsevier)

The temperature measured in situ was lower than 45°C in all cases. It is observed that HiPIMS gives better resistivity values than DC for these specific conditions and that increasing the discharge voltage in HiPIMS further lowers the resistivity and improves strongly its homogeneity as the peak facing the target axis tends to disappear. Using 570 V, very low resistivity values of 4-15 10^{-4} Ωcm depending on position could be achieved while keeping a good transparency in the visible range. This is of strong interest for the development of transparent electrodes used as single layer or component of devices to be synthesized on organic supports or containing materials degrading at moderate temperature. The average optical transmittance in the visible range (380-780 nm) ranged from 72 to 86 % for these samples, generally rising with the resistivity.

We found that the more conductive films show the lowest crystallinity, in contrast with most reports on the electrical properties of AZO films that correlate good electrical properties to optimized crystallinity. The good results presented here are due to a large density of electrons in the range 5-11 10^{20} cm^{-3} depending on position. At such high electron densities, grain boundary scattering weakens due to lowered potential barrier [1]. Scattering by ionized impurities becomes the dominating limiting factor to the electron mobility. Several measurements indicated oxygen sub-stoichiometry inside the grains. Hence, donor defects should be dominant. A slight oxygen sub-stoichiometry in AZO films seems to be important to obtain large conductivity but this is probably not the only condition as was discussed above. Indeed, as many as possible aluminum atoms must be activated and the presence of acceptor defects that could compensate the donor ones should be avoided. The density of atoms sputtered from the target during a pulse using our HiPIMS conditions is similar to the surface atom density of ZnO. This means, even if the target surface oxidizes between pulses, it can be cleaned during pulses. The discharge voltage regulates the sputtering rate and, thereby, the ability to clean the target. This fully explains the results of figure 6. Some studies tend to indicate the possibility to achieve similar effects during reactive HiPIMS. Ganesan et al. also explained recently their results of varied oxygen composition during reactive HiPIMS deposition of HfO_2 and NbO_2 [29, 30]. Finally, Wallin and Helmersson have reported in 2008 on the possibility to deposit alumina layers without the hysteresis effect usually observed in highly reactive systems such as Al-O [31].

Target cleaning limits the bombardment by fast oxygen particles that generate acceptor defects in ZnO as stated in the previous section. Figure 7 (up) shows the Al K edge XANES spectra for AZO films deposited in optimized conditions in the same reactor at 0 (solid lines) and 30 mm (dash lines) from the target axis and using direct current, HiPIMS and medium frequency (MF) sputtering. The heterogeneity of electronic structure is obvious for DC and MF while hardly detected for HiPIMS. Figure 7 (down) links the inhomogeneity of conductivity to that of the aluminum dopant activation. It is observed that the ratio from maximum to minimum conductivity measured at different positions scales very well with the intensity ratio of the CN6 contribution in-axis and off the magnetron axis in XANES at the Al K edge spectra, which highlights the importance of the dopant activation. Nevertheless, it is difficult to disentangle the relative influences of donor/acceptor intrinsic defects and dopant activation/deactivation because both may be strongly affected by the magnitude of the bombardment by energetic particles. Moreover, the density of electrically active defects is hardly accessible due to the far out of stable

equilibrium condition associated to the condensation of the gas phase and ballistic effects.

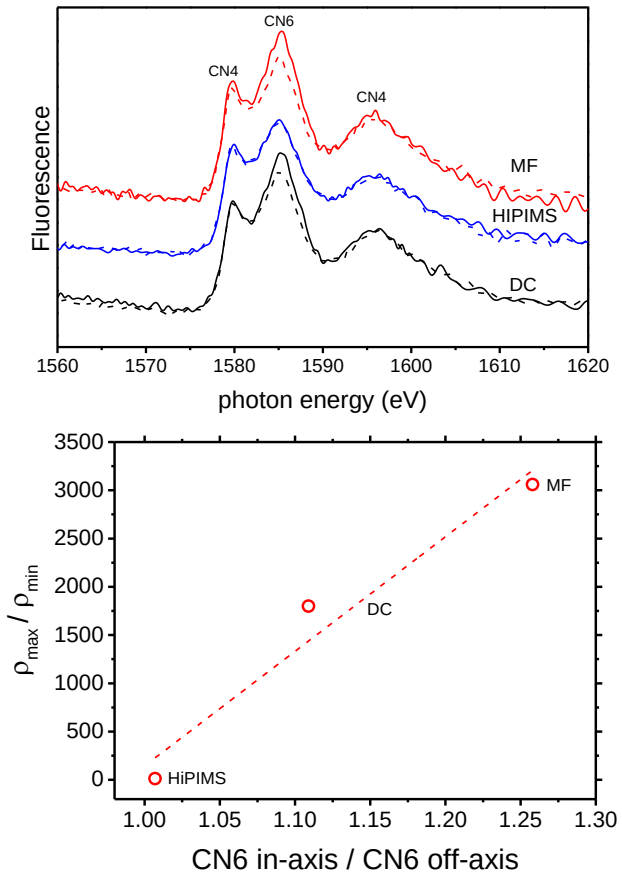


Figure 7 (up) Total fluorescence yield Al K edge XANES spectra of AZO thin films deposited at circumferences of 0 and 30mm from the target axis using HiPIMS, direct current (DC) and medium frequency (MF) reactive sputtering. (down) Evolution of the intensity ratio of maximum (30mm) to minimum (0 mm) resistivity as a function of the intensity ratio of CN6 at 0 mm to CN6 at 30 mm.

Finally, we present on figure 8 the PL spectra for the samples deposited at 4 and 8 cm from the substrate holder axis at 570 V using HiPIMS (see figure 6). It reveals a very small signal in the visible range, that might be indicative of a low density of point defects and a strong emission in the UV due to recombination of free excitons [28]. The intensity ratio between the UV and visible contributions is often used as a criterium to evaluate the crystallinity of ZnO. This could suggest little perturbation of the structure inside the grains of our AZO films. Nevertheless, high resolution transmission electron microscopy (HRTEM) analyses showed the grains are small with size close to 20 nm and X-ray diffraction indicated a lower crystallinity than for DC films [28]. Hence, the validity of this criterium is questionable.

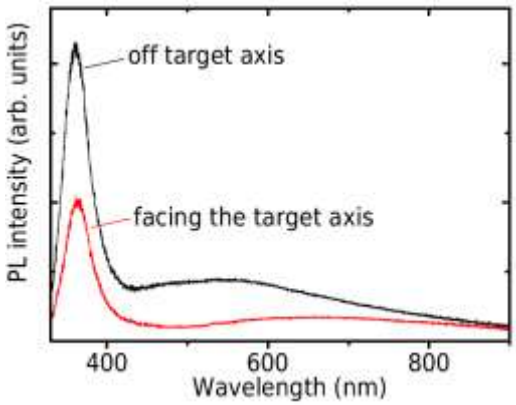


Figure 8 Photoluminescence spectra of a HiPIMS deposited AZO film for different positions (reprint from [28], copyright elsevier)

3 Conclusion We have shown that reactive magnetron sputtering can be successfully employed to manipulate the microstructure and electrical properties of ZnO films without thermal assistance during or after deposition. In a first part we have described the epitaxy at room temperature and proposed that it can be achieved by fast oxygen particles impinging the growing film and transferring their energy to the film. In a second section, we have first briefly reviewed the limitation of DC reactive sputtering to establish simultaneously highly transparent and conductive aluminum-doped ZnO films on large surface areas. Finally, we have described a new method based on reactive HiPIMS we reported recently that allows achieving this objective. The results open new horizons for the use and study of ZnO based coatings in the fields of optoelectronics and flexible electronics.

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